

Notes to paperes

Some subtitle for my thesis

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Abstract

The three-dimensional organization of the genome plays a fundamental role in gene regulation, evolution, and germ cell development. Chromosomes are partitioned into transcriptionally active A-compartments and repressive B-compartments, with boundaries between them representing dynamic sites of chromatin reorganization. This thesis investigates the sequence determinants underlying these compartment boundaries in *Macaca* genomes, focusing on spermatogenesis and the distinctive regulatory behavior of the X chromosome. Initially, this study aimed to leverage Genome Pretrained Networks (GPNs) to predict chromatin states and evolutionary conservation from DNA sequence data. While conceptually promising, this approach proved technically challenging due to inadequate documentation of published workflows and significant hardware limitations. Even with multi-GPU configurations, full-genome training remained computationally infeasible, constraining analyses to chromosome-level embeddings. These obstacles underscore the practical difficulties of implementing cutting-edge deep learning methods in genomics research within typical academic resource constraints. The study therefore pivoted to classical bioinformatic approaches, examining sequence features at A/B compartment boundaries across five spermatogenic developmental stages. GC content and CpG islands emerged as strong predictors of compartment identity on autosomes, exemplified by chromosome 8, but these correlations were notably weakened or reversed on the X chromosome, reflecting its unique regulatory landscape. CpG islands demonstrated dynamic, stage-specific enrichment patterns in A-compartments during later spermatogenic stages, coinciding with periods of extensive chromatin remodeling. Repetitive elements exhibited contrasting distribution patterns: enriched in B-compartments on autosomes but preferentially located in A compartments on the X chromosome. Importantly, repetitive elements consistently clustered near compartment boundaries regardless of chromosomal context, suggesting potential structural or regulatory functions in chromatin organization. These findings demonstrate that while autosomes generally conform to established sequence-compartment relationships, the X chromosome exhibits distinct organizational principles that deviate from canonical patterns. This work establishes a framework for integrating sequence composition, repetitive elements, and chromatin architecture, while highlighting the critical importance of considering both biological context and methodological constraints when interpreting three-dimensional genome organization.

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1 Introduction

The central aim of this thesis is to investigate chromosomal regions where chromatin transitions between A- and B-compartments, specifically focusing on the sequence features and regulatory elements that accompany these boundaries. These transitions are of particular interest because they represent dynamic sites of genome reorganization, linking sequence composition to higher-order chromatin structure and, ultimately, to functional outcomes in gene regulation and evolution.

The original idea for this project was to leverage artificial intelligence—specifically genomic pre-trained networks (GPNs)—to predict conservation across species and compare these predictions with chromatin compartment organization. While technically ambitious, this initial approach proved infeasible within the scope of the project, primarily due to computational limitations and the complexity of reproducing large-scale models. Rather than abandoning the question, the project pivoted toward more classical and tractable methodologies, extending the investigation of chromatin transitions through established bioinformatic and comparative genomic approaches.

This transition in strategy opened up the possibility of exploring compartment boundaries from multiple complementary angles: analyzing sequence features such as GC content, CpG islands, and repetitive elements; assessing their distribution relative to compartment edges; and considering their potential roles in genome function and evolution. In doing so, the work contributes both methodological insights—on the challenges of applying AI in genomics—and biological insights into how compartmental transitions are structured and maintained.

1.1 evolution on X

The production of gametes in sexually reproducing organisms is a complex, tightly regulated process involving numerous cellular and molecular mechanisms. In males mammals, spermatogenesis—the process by which sperm are formed—progresses through four main stages of differentiation: from germ cells to spermatogonia, then to pachytene spermatocytes, round spermatids, and finally to mature spermatozoa (Wang et al. 2019). This process is fundamental to male fertility. Meiosis, a specialized form of cell division, ensures the proper pairing, recombination, and segregation of homologous chromosomes, thereby maintaining genetic integrity and promoting variation. A deep understanding of these molecular events is essential for comprehending inheritance, speciation, population genetics, and the biological causes of infertility (Wang et al. 2019).

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Sex chromosomes differ from autosomes in several key ways due to their distinct patterns of inheritance and copy number. The Y chromosome is found only in males and exists as a single copy. In contrast, the X chromosome follows a more complex inheritance pattern: females carry two copies, while males carry only one. Consequently, the X chromosome resides two-thirds of the time in females and one-third in males, a distribution that influences how selection acts upon it. In males, the hemizygous nature of the X chromosome means that any mutations or deleterious alleles are fully exposed, without a second copy to mask their effects (Abe et al. 2020).

This feature is central to Haldane’s rule (Haldane 1922), which posits that in hybrids between two species, the heterogametic sex (e.g., XY in mammals or ZW in birds) is more likely to be inviable or sterile. While widely observed, the precise causes of this pattern remain a subject of ongoing debate. Several hypotheses have been proposed to explain it. One suggests that incompatibilities involving the Y chromosome can arise if it fails to remain compatible with the X chromosome or autosomes (Haldane 1922). Another centers on dosage compensation, proposing that hybridization may disrupt the regulatory mechanisms that balance gene expression between sex chromosomes. The dominance hypothesis argues that recessive deleterious alleles may be unmasked in the heterogametic sex, leading to sterility. Additional ideas include the faster evolution of male-biased reproductive genes, which can result in functional mismatches, and the accelerated divergence of X-linked loci compared to autosomes, known as faster-X evolution. Lastly, meiotic drive—genetic conflicts between selfish elements on sex chromosomes—has also been implicated in hybrid sterility.

These hypotheses underscore the complex interplay between sex chromosomes, selection, and hybridization (Cowell 2023). Although Haldane’s rule is consistently observed across diverse taxa—and even across kingdoms—the underlying mechanisms are not universally agreed upon. In many cases, multiple processes may act together (Lindholm et al. 2016).

The X chromosome, in particular, appears to be subject to strong and unique selective pressures, especially in primates. This topic is explored in greater depth in the following sections.

1.2 Chromatin structure

Regions under strong selection across primate species—including humans and baboons—are consistently found on the X chromosome, spanning megabase-scale genomic areas. As previously noted, factors such as chromosomal architecture and structural rearrangements may play crucial roles in enabling, regulating, or insulating these regions from selective pressures.

Understanding genome organization and variation is therefore essential for uncovering the mechanisms behind selection. Chromatin has long been recognized as central to gene regulation and cellular function (Lieberman-Aiden et al. 2009) (Lieberman-Aiden et al. 2009), in part because the 3D structure of chromosomes brings distant genomic elements into close contact. Disruption of this spatial organization can lead to developmental abnormalities (Dixon et al. 2015) (Dixon et al. 2015). Chromatin is organized hierarchically, with multiple levels of structure

nested within one another. At the broadest level, chromatin compartments can span several megabases (Lieberman-Aiden et al. 2009) (Lieberman-Aiden et al. 2009), while at finer scales, even 500 base pairs can contribute to subgenomic structural organization. Structures such as topologically associating domains (TADs) and chromatin loops, which operate at sub-megabase scales, further help partition regulatory elements and maintain proper genomic function [Ramírez et al. (2018)](Zuo et al. 2021)(Ramírez et al. 2018; Zuo et al. 2021).

Chromatin, the complex of DNA wrapped around histone proteins and associated with various regulatory factors, plays a central role in the spatial and temporal control of gene expression, as well as in maintaining the structural integrity and stability of the genome (Allshire and Madhani 2016; Bannister and Kouzarides 2011). Far from being a static scaffold, chromatin is a highly dynamic entity, subject to extensive remodeling in response to cellular signals, developmental cues, and environmental stimuli (Maeshima et al. 2020). Its organization within the nucleus is hierarchical, ranging from nucleosomes to higher-order domains, and this architecture can differ substantially across cell types, developmental stages, and even between closely related species (Rowley and Corces 2018; B. Bonev and Cavalli 2017).

One of the key features of chromatin organization is its division into functionally distinct states—broadly classified as euchromatin, which is generally open and transcriptionally active, and heterochromatin, which is compacted and repressive (Grewal and Jia 2007; Bickmore and Steensel 2013). Transitions between these states, whether developmentally programmed or evolutionarily emergent, are often tightly coupled to shifts in gene regulatory activity and cellular identity (Arnold 2022).

this study aims to characterize these chromatin state transitions, with a particular interest in understanding how they are distributed across the genome and whether they exhibit predictable patterns linked to specific sequence features or regulatory elements. This study examines whether certain genomic regions are more prone to undergo structural reorganization and whether such regions harbor conserved sequence motifs, epigenetic signatures, or binding sites for key regulatory proteins (Zhou et al. 2011; Ernst and Kellis 2011). Furthermore, this study investigates whether particular chromatin configurations are disproportionately associated with genes involved in critical biological processes or with regions of the genome that show signatures of evolutionary constraint or adaptation (Kundaje et al. 2015).

By examining the diversity of chromatin states and transitions across individuals and lineages, this study also aims to uncover broader principles governing chromatin architecture and its functional consequences (Marx 2019). These insights may inform our understanding of how chromatin organization evolves, how it contributes to phenotypic diversity, and how its dysregulation may underlie disease states (Buenrostro et al. 2015). Ultimately, this work contributes to a growing body of knowledge on the interplay between genome structure, function, and evolution.

1.3 A/B configurations

One of the most fundamental and widely studied features of three-dimensional genome organization is its partitioning into A and B-compartments, originally revealed by high-throughput chromosome conformation capture techniques such as Hi-C (Lieberman-Aiden et al. 2009). These compartments represent megabase-scale regions of the genome that differ significantly in their epigenetic state, transcriptional activity, nuclear positioning, and physical interaction profiles. The identification of compartments via principal component analysis (PCA) of Hi-C matrices has become a cornerstone in mapping chromatin architecture across species and cell types.

The A-compartment is typically associated with euchromatin, that supports active gene expression. Regions within A-compartments are enriched with transcriptionally active histone modifications, including H3K27ac, H3K4me3, and H3K36me3, as well as high levels of DNase I hypersensitivity, RNA polymerase II occupancy, and active transcription (Han, M.H., Issagulova, D., and Park, M. 2023; Wen, Z. et al. 2022; Lu, J.Y. et al. 2021). In contrast, B-compartments correspond to heterochromatic domains, which are transcriptionally silent or repressed. These regions are often localized near the nuclear lamina, marked by H3K9me3, H3K27me3, Lamin B1 association, and DNA methylation, and exhibit low transcriptional activity (Han, M.H., Issagulova, D., and Park, M. 2023; Lu, J.Y. et al. 2021; Feng, Y. et al. 2020). The biochemical and physical segregation of chromatin into A and B-compartments reinforces a higher-order regulatory framework for genome function (Lucini, F. et al. 2024).

Importantly, A/B compartment identity is not fixed but can undergo dynamic switching in response to developmental cues, environmental stimuli, cell differentiation, aging, or disease states such as cancer and senescence. This process, referred to as compartment switching, entails a region transitioning from a repressive B-compartment to an active A-compartment (or vice versa), often accompanied by changes in chromatin accessibility, histone modification landscapes, and gene expression profiles (Zhang, X. et al. 2021; Lucini, F. et al. 2024; Wen, Z. et al. 2022). For instance, during lineage specification or upon activation of certain signaling pathways, genes or regulatory loci may shift from a transcriptionally repressive nuclear environment to one permissive for gene activation. This spatial repositioning allows cells to coordinate complex transcriptional programs in a spatiotemporally regulated manner.

At the structural level, compartmental identity correlates with nuclear topography: A-compartments tend to localize to the nuclear interior and are often associated with nuclear speckles—subnuclear domains involved in transcriptional and RNA processing. B-compartments, on the other hand, are frequently found near the nuclear lamina, where gene silencing is promoted through anchoring and phase-separated heterochromatin domains (Lu, J.Y. et al. 2021; Feng, Y. et al. 2020).

From an evolutionary perspective, comparative Hi-C and epigenomic studies across species have revealed that certain A and B compartmental domains are highly conserved, especially those involving developmental genes, housekeeping genes, and essential regulatory networks (Valadez-Graham, V. and Magaña-Acosta, M. 2020). However, other regions—particularly those involved

in immune response, sensory perception, or species-specific traits—exhibit compartmental plasticity, reflecting their role in adaptive evolution and phenotypic diversity. For example, chromatin compartments in plants, mammals, and even non-model organisms show patterns of both conservation and innovation in their compartmental structures, highlighting the flexibility of genome organization mechanisms in accommodating species-specific regulatory needs (Wen, Z. et al. 2022).

The study of A/B compartment dynamics not only offers insights into basic principles of genome function but also provides a framework for understanding disease-associated epigenetic reprogramming, evolutionary divergence, and the mechanistic underpinnings of cell identity.

1.4 GPN

Recent advances in artificial intelligence have brought powerful tools to genomics, particularly through the development of Genomic Pre-trained Networks (GPNs)—deep learning models trained on large-scale genomic sequence data using architectures inspired by natural language processing. GPNs apply self-supervised learning approaches, analogous to models like BERT or GPT, to the DNA sequence, learning to predict masked or missing bases in context. Through this training, GPNs develop an internal representation of the “language of the genome,” capturing patterns of sequence composition, motif syntax, and long-range dependencies that underlie regulatory function (Ji et al. 2021; Cheng, Wang, and Kundaje 2021; Sasse, Mukamel, and Kundaje 2023; N. Benegas et al. 2023).

A key advantage of GPNs lies in their use of transformer architectures, which differ significantly from traditional models like convolutional neural networks (CNNs) (Kelley, Snoek, and Rinn 2016). While CNNs are adept at recognizing short, fixed-length patterns—such as transcription factor binding motifs—they struggle with learning dependencies over long genomic distances, such as enhancer-promoter interactions or 3D chromatin looping (Avsec et al. 2021). Transformers, in contrast, leverage self-attention mechanisms, which allow the model to compare and weigh relationships between all nucleotide positions in a sequence, regardless of distance. This makes them exceptionally well-suited for modeling the complex, multi-scale structure of regulatory DNA (N. Benegas et al. 2023).

The utility of GPNs in functional genomics has been demonstrated in tasks such as predicting the effects of noncoding variants, chromatin accessibility, and transcription factor binding. For instance, the model introduced by Benegas and Batra (N. Benegas et al. 2023) (“DNA language models are powerful predictors of genome-wide variant effects”) showed that transformer-based GPNs outperform prior models in predicting variant impact across a variety of cell types and epigenomic contexts, even without explicit annotations. These models can generate base-resolution functional predictions across the genome, enabling insight into the likely regulatory role of any given sequence segment.

In the context of chromatin structure, GPNs hold particular promise for identifying sequence features that influence or correlate with A/B compartment identity. By learning patterns that

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distinguish sequences in open (A) versus closed (B) chromatin, GPNs can be used to predict how shifts in compartmentalization may arise from sequence variation. Furthermore, these predictions can be extended across species or populations to assess functional conservation—that is, whether a sequence maintains regulatory potential or compartment association despite genetic divergence.

This study leverages GPN-derived embeddings and predictions to investigate the sequence determinants of chromatin organization. This study aims to identify genomic regions where the GPN model predicts strong, conserved regulatory activity, and determine whether these regions also show structural stability in chromatin compartments across individuals or hybrid genomes. This approach allows us to assess whether chromatin transitions are driven primarily by changes in DNA sequence, or whether other regulatory layers (e.g., epigenetic modifications, chromatin remodelers) are required to explain shifts between A and B compartments.

By integrating Genomic Pre-trained Network (GPN) predictions with chromatin conformation data such as Hi-C, researchers can construct a powerful multiscale framework for understanding genome function—linking base-pair regulatory logic with higher-order nuclear architecture (N. Benegas et al. 2023; Schultz, Kyhl, and Willett 2025). These models not only enhance the interpretation of noncoding variation and functional conservation but also offer a means to study how genomic sequence encodes three-dimensional chromatin organization. In systems such as primates, where population structure and hybridization events are prevalent, this approach opens new avenues for investigating the evolution of regulatory architecture in action (Eres 2020; Álvarez González 2024).

1.5 Meiosis

Meiosis is the defining process of gametogenesis, transforming diploid precursors into haploid gametes while introducing genetic diversity. The stages represented in this dataset highlight distinct steps in this transformation (Berg et al. 2019).

Spermatogonia are the mitotically active germline stem and progenitor cells at the basal compartment of the seminiferous tubules. Although they themselves do not undergo meiosis, they form the essential starting population that differentiates into spermatocytes, committing cells to the meiotic program (Berg et al. 2019).

Pachytene spermatocytes represent a central point in meiosis, specifically prophase I. During this prolonged phase, homologous chromosomes are fully synapsed through the synaptonemal complex, and crossing-over ensures genetic recombination. This is a transcriptionally active stage, producing RNAs that will later support spermatid development when transcription decreases (Berg et al. 2019).

Round spermatids are the direct products of the two meiotic divisions. They are haploid cells that no longer divide but instead undergo spermiogenesis, a process of dramatic morphological

and chromatin remodeling. Although post-meiotic, they carry the molecular signatures of meiotic recombination and genome reduction (Berg et al. 2019).

During spermatogenesis, large-scale reprogramming of three-dimensional (3D) chromatin organization is required to enable meiosis and germ cell differentiation. Recent studies using low-input Hi-C in both rhesus monkeys and mice have revealed that the higher-order chromatin landscape is far from static. Instead, it undergoes stage-specific remodeling in which topologically associating domains (TADs), normally stable units of genome folding, become progressively dissolved as germ cells enter meiotic prophase. This disruption reaches its peak in pachytene spermatocytes, where TADs are almost entirely absent despite the presence of widespread transcriptional activity (Wang et al. 2019).

Interestingly, the loss of TADs at this stage is accompanied by the emergence of refined A/B compartments. These compartments alternate between transcriptionally active and inactive regions, providing a new level of chromatin organization that correlates more directly with gene expression than classical TAD structures. Such reorganization suggests that the meiotic genome adopts a unique 3D folding principle optimized for homologous recombination and transcriptional control (Wang et al. 2019).

A key driver of this transition appears to be the synaptonemal complex (SC), a protein scaffold that aligns homologous chromosomes during meiosis. Wang and colleagues demonstrated that in spermatocytes lacking proper SC formation, refined compartments fail to develop, and TAD-like structures re-emerge prematurely. This finding indicates that the SC does not merely stabilize chromosome pairing, but also actively contributes to global chromatin architecture by suppressing TADs and promoting compartmentalization (Wang et al. 2019).

Overall, this work highlights that spermatogenesis involves not only dynamic transcriptional and epigenetic changes but also profound reprogramming of chromatin folding. The discovery that such architectural shifts are conserved between primates and rodents underscores their biological importance. Furthermore, it raises the possibility that defects in these processes could contribute to infertility or other germline disorders, linking 3D genome organization to reproductive health (Wang et al. 2019).

1.6 X Chromosome Compartments and GC Content

One of the defining molecular features of A/B chromatin compartments is their GC content, which reflects and reinforces their functional states. A-compartments are generally GC-rich and correspond to gene-dense, euchromatic regions. These areas are transcriptionally active, marked by high RNA polymerase II occupancy and enriched in active histone modifications such as H3K4me3 and H3K27ac. Their open chromatin conformation also renders them accessible to DNase I and other chromatin accessibility assays [Lieberman-Aiden et al. (2009)] (Boyan Bonev and Cavalli 2016).

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In contrast, B-compartments tend to be AT-rich, gene-poor, and heterochromatic. These regions are associated with repressive histone marks like H3K9me3, delayed replication timing, and frequent localization near the nuclear periphery, particularly within lamina-associated domains (LADs). The compositional contrast between A and B-compartments is not only a reflection of gene content but also contributes to chromatin folding, nucleosome positioning, and transcription factor binding patterns (Lieberman-Aiden et al. 2009).

Importantly, GC content remains a stable genomic feature, even as compartment identity can shift in response to developmental or environmental changes. For example, during cell differentiation or meiotic prophase, such as the pachytene stage, there is widespread chromatin reorganization, and architectural domains like TADs may temporarily dissolve. Yet, GC-rich regions consistently align with active compartments, suggesting that GC content serves as an intrinsic guide for nuclear organization, independent of dynamic epigenetic changes (Wang et al. 2019).

This stability makes GC content a robust predictor of A/B compartment identity across cell types and developmental stages, underscoring its fundamental role in genome architecture (Wang et al. 2019).

1.7 CPG islands

CpG islands (CGIs) are typically unmethylated DNA regions that are closely associated with gene promoters and play a crucial role in regulating chromatin accessibility and three-dimensional genome organization. Accumulating evidence demonstrates that changes in CGI methylation, histone modifications, or chromatin-binding proteins can drive or accompany transitions between chromatin compartments, shifting from active A-compartments to repressive B-compartments or vice versa. Such transitions are frequently observed in developmental processes, cancer progression, X-chromosome inactivation, and transcriptional regulation. In many cases, CGIs act as anchors or boundaries that help define chromatin domains, and their epigenetic state can strongly influence large-scale compartmental reorganization.

Recent research on meiotic chromatin architecture provides insights that support this broader view of CGI function. A study by Wang *et al.* (2019) (Wang et al. 2019) examined spermatogenesis in rhesus monkeys and mice using low-input Hi-C and found that canonical TADs dissolve during the pachytene stage of meiosis, even while transcription continues. This observation suggests that TAD integrity is not strictly required for gene expression at this stage. Instead of the standard large-scale A/B compartmentalization, the study identified the emergence of smaller, refined A/B compartments that alternate between transcriptionally active and inactive domains. The formation of these fine-scale compartments was shown to depend on the synaptonemal complex (SC). In SC-defective mutants, canonical TADs reappeared, indicating that the SC suppresses TADs while enabling this unique compartmental structure. Notably, this reorganization of 3D chromatin architecture was conserved between monkeys and mice, suggesting a mammalian-wide regulatory principle (Wang *et al.*, 2019) (Wang et al. 2019).

Although CpG islands were not the central focus of this study, the findings are consistent with the hypothesis that CGIs contribute to compartmental organization by functioning as boundary or anchoring elements. Because refined compartment transitions align with transcriptional activity, and CGIs are commonly associated with active transcription start sites, it is likely that they play a role in stabilizing the boundaries between alternating compartments. In this way, CGIs may provide local chromatin insulation while also influencing higher-order genome organization.

1.8 repeat

1.8.1 simple nested repeats

Repetitive elements make up a large portion of the eukaryotic genome and are broadly categorized into simple (tandem) and nested (interspersed) repeats. Simple repeats, such as microsatellites, minisatellites, and satellite DNA, consist of short motifs repeated in tandem and are frequently found in centromeric, subtelomeric, and regulatory regions. They are prone to replication slippage and are widely used in genetic mapping and disease diagnostics due to their high mutation rates (Gemayel et al. 2010) . In contrast, nested repeats, derived largely from transposable elements (TEs) such as LINEs, SINEs, and LTR retrotransposons, are scattered throughout the genome. These elements can insert into one another, creating complex repeat-rich regions that influence genome structure, epigenetic regulation, and chromatin organization (Bourque et al. 2018).

2 Materials and Methods

2.1 Genome-Pretrained Network (GPN) architecture

The idea for this project was to evaluate the GPN introduced in G. Benegas, Batra, and Song (2024) and determine whether it could achieve greater accuracy than traditional methods in predicting genome-wide variant effects.

The model was designed as a convolutional neural network (CNN) and took input sequences with a window size of 512. During training, 15% of the positions within each window were masked to enable the model to learn meaningful representations.

The architecture consisted of 25 layers, each structured as follows: a dilated convolution layer, followed by an add-and-norm layer with a skip connection from before the dilated convolution. This was followed by a feedforward layer, another add-and-norm layer, and additional skip connections.

A feedforward layer was a fundamental component of neural networks, where inputs passed through one or more fully connected layers with activation functions, transforming data without looping back. This structure helps the model learn complex representations by applying weighted transformations and non-linearities G. Benegas, Batra, and Song (2024).

A dilated convolution expands the receptive field of a convolutional layer without increasing the number of parameters or reducing resolution. By spacing kernel elements apart, it captured long-range dependencies in sequences, making it particularly useful in genomic data analysis. When combined, dilated convolutions and feedforward layers enhanced a model's ability to recognize patterns across different scales efficiently.

After passing through the 25 layers, the model produced a contextual embedding with a dimension of 512 ($D=512$), followed by classification layers. The final layer outputs the probabilities of the four nucleotides at each masked position.

The GPN variant effect prediction score was calculated as the log-likelihood ratio between the alternate (ALT) and reference (REF) alleles. Here, L represents the window length in base pairs, and D denotes the embedding dimension.

2.1 Genome-Pretrained Network (GPN) architecture

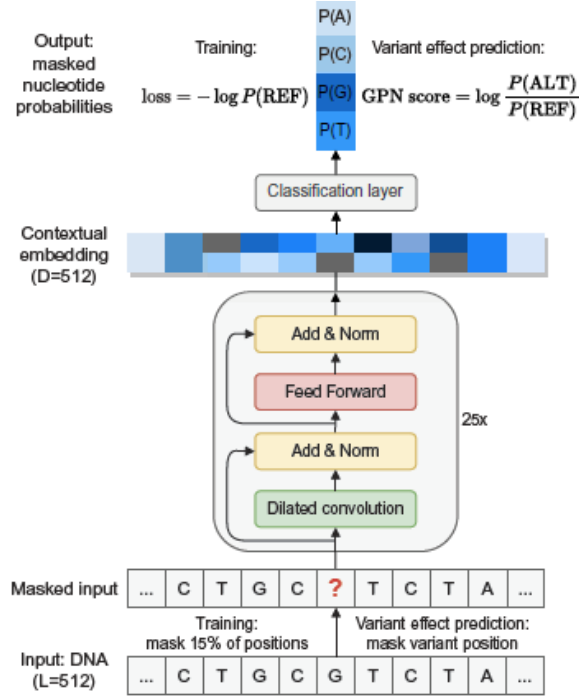


Figure 2.1: **Figure 1.** Schematic overview of the Genome-Pretrained Network (GPN) architecture, adapted from G. Benegas, Batra, and Song (2024). The model processes genomic sequences using stacked dilated convolutional and feedforward layers with skip connections, producing contextual embeddings that support masked nucleotide prediction and variant effect scoring.

2.2 Data Access

We based our implementation work on the official **GPN** repository maintained by Song Lab: <https://github.com/songlab-cal/gpn>

In particular, we referred to the **analysis/gpn_arabidopsis** subdirectory of the repository as an example of end-to-end model application on a plant genome: https://github.com/songlab-cal/gpn/tree/main/analysis/gpn_arabidopsis.

2.3 Data Preparation and Pretraining Workflow

To evaluate the GPN, the initial step involved reproducing the pretraining workflow provided in the original publication G. Benegas, Batra, and Song (2024). This workflow outlines the procedure for generating a dataset compatible with the model, specifically for masked language modeling over genomic sequences.

A dataset suitable for GPN pretraining was successfully generated. The resulting data consisted of masked genomic sequences, formatted according to the model’s input requirements.

2.4 Model Execution Workflow and Hardware Constraints

The step after the successful generation of a pretraining-compatible dataset involved attempting to run the GPN using the available inference and training workflows. The documentation provided on the model’s GitHub repository lacked detailed guidance.

To proceed, an alternative workflow was identified, **analysis/gpn_arabidopsis** specifically designed for use with the *Anatoptis* genome. This workflow was adopted as a reference implementation for executing the GPN model.

However, significant hardware limitations emerged during this phase. The original study reported using four NVIDIA A100 GPUs, each equipped with 80 GB of RAM. In contrast, the available system utilized NVIDIA L43R GPUs with 45 GB of RAM, resulting in substantial differences in computational capacity.

As a result, attempts to execute the GPN model encountered frequent out-of-memory errors, particularly during forward passes through the deeper layers of the network. These memory constraints posed a major obstacle and significantly hindered reproducibility under the available hardware setup.

Initial efforts to train or fine-tune the GPN focused on applying the workflow to a baboon genome using a multi-GPU setup. The goal was to replicate the original training environment, which employed four NVIDIA A100 GPUs (80 GB RAM each). However, the available infrastructure

utilized NVIDIA L43R GPUs, each with 45 GB of RAM, which introduced several compatibility and performance challenges.

Early testing began with the Anatoptis genome. Attempts to process the entire genome using the Snakemake workflow were unsuccessful due to GPU detection issues. These issues stemmed from the cluster configuration, which prevented the Snakemake environment from correctly interfacing with the available GPU hardware.

After resolving these initial configuration issues, the workflow was executed with GPU support. While this improved performance, the model began encountering memory errors after a few hours of training, even when utilizing two to four GPUs in parallel. In an effort to overcome these constraints, the number of GPUs was incrementally increased—from four to six, and finally to eight providing an aggregate of 360 GB of GPU RAM.

Ultimately, these constraints prevented successful execution of the GPN model on the tested system.

Given the persistent memory limitations encountered during full-genome processing, an alternative strategy was implemented. Rather than inputting the entire Anatoptis genome, the data was partitioned at the chromosome level. This allowed individual chromosomes to be processed independently, while still allocating the full available GPU resources (eight NVIDIA GPUs with a combined 360 GB of RAM) to each run.

This approach proved successful. When processing a single chromosome at a time, the GPN model was able to execute without encountering memory-related failures. As a result, contextual embeddings were successfully generated for individual chromosomes of the Anatoptis genome.

However, due to time constraints near the conclusion of the project timeline, the extracted embeddings were not subjected to further downstream analysis. Although they were successfully retrieved and stored, no additional modeling, interpretation, or evaluation steps were completed within the project scope.

2.5 Fecting the raw data Hi-c analysis

The original sequencing data was obtained by a colleague in our research group in order to reproduce the results reported by (Wang et al. 2019). The raw data was downloaded from the SRA portal (Sayers et al. 2022) and subsequently filtered to include only paired-end Hi-C reads from macaque samples.

To provide an overview of the material, a summary of the SRA runtable is presented in Table 2.1. The accession numbers alongside key metadata for each sample are listed in the table. Altogether, the dataset contained 1 TB of compressed FASTQ files, comprised ~9.5 billion reads distributed roughly evenly across five tissue types.

Source_name	GB	Bases	Reads
Fibroblast	211.403275	553,968,406,500	1,846,561,355
Pachytene spermatocyte	274.835160	715,656,614,700	2,385,522,049
Round spermatid	243.128044	655,938,457,200	2,186,461,524
Sperm	164.131640	428,913,635,400	1,429,712,118
Spermatogonia	192.794420	518,665,980,300	1,728,886,601

Table 2.1: Summary of Hi-C sequencing data across five macaque cell types, retrieved from the NCBI SRA portal for initial data exploration.

2.5.1 Data Generation and Processing

The Hi-C sequencing data analyzed in this study was not generated or processed by the author, but by another member of our research group. To enable clarity and reproducibility, the following section contains a condensed description of their workflow.

The data were processed using the Open2C pipeline, implemented as a GWF workflow. This framework allowed each sequencing accession (i.e., paired-end read sets) to be handled independently and in parallel, improving efficiency and scalability. The raw reads were retrieved directly from the NCBI SRA (DevTeam, SRA Toolkit 2024) using the SRA Toolkit, run in a containerized environment for consistency. Downloads were saved as compressed FASTQ files, ensuring a standardized input format for downstream analysis.

Read alignment was carried out against the rheMac10 reference genome using bwa-mem in paired-end mode, following the recommended parameters provided by the Open2C team. The resulting alignments were then parsed into ligation events with pairtools, which not only distinguishes valid Hi-C contacts but also applies essential preprocessing steps such as sorting, removal of PCR duplicates, and exclusion of reads mapping to unplaced scaffolds. This filtering ensured that only informative, high-confidence pairs were retained for analysis.

The resulting valid pairs were aggregated into cooler files, the standard data structure for Hi-C contact matrices. To maximize coverage, biological replicates were merged at the cell-type level into a single matrix. These contact matrices were then “zoomified” into multiple resolutions (10 kb to 500 kb bins), enabling both fine-scale and large-scale chromatin features to be studied within the same dataset. To correct for technical biases, each matrix was subsequently normalized using iterative correction, producing balanced matrices suitable for quantitative comparison.

To identify higher-order chromatin organization, eigendecomposition of the contact matrices was performed. The first eigenvector (E1) was phased using GC content, ensuring consistent orientation of A/B compartments across chromosomes. While eigenvectors were computed at multiple resolutions during intermediate steps, all final analyses and compartment calls were performed at a uniform resolution of 100 kilobases (kb) to ensure consistency across datasets.

The eigentrack shown in Figure 2.2 illustrates a representative example of the first eigenvector at this resolution.

Finally, visualization of contact matrices and compartments was conducted within the Open2C ecosystem using matplotlib and seaborn, which enabled efficient data access and flexible plotting. These tools allowed for side-by-side comparison of balanced versus unbalanced matrices, as well as detailed compartment tracks at 100 kb resolution.

Together, this workflow produced a high-quality, reproducible dataset. The combination of containerized processing, rigorous filtering, normalization, and replicate merging ensured that the resulting chromatin contact maps were reliable and well-suited for downstream compartment analysis.

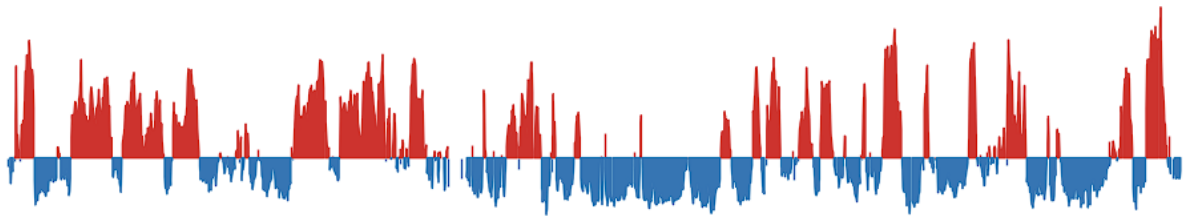


Figure 2.2: **Example of the first eigenvector (“eigentrack”) used in A/B chromatin compartment analysis.**

This track represents the first principal component (PC1) of the Hi-C contact matrix at a resolution of 100 kilobases (kb), with eigenvector values ranging from -1 to $+1$. Genomic regions with positive eigenvector values (colored red) are assigned to A-compartments, typically associated with open, transcriptionally active euchromatin. Conversely, negative values (colored blue) correspond to B-compartments, generally representing condensed, transcriptionally repressive heterochromatin. This eigentrack provides a quantitative basis for segmenting the genome into large-scale chromatin domains based on spatial interaction patterns.

2.6 Datahandeling before the analysis

To load the data the function `parse_compartment_data(file_name)` was used, which transforms the raw eigenvector track into meaningful genomic segments. The function began by reading a CSV file that contains at least one column, `e1`, representing eigenvector values at 100 kb resolution.

The eigenvector track (E1) marked A/B compartments along the genome. During its generation, some bins may contained NaN values. If these were left in the dataset, each missing value was treated as a break and produces false edges, creating compartment transitions that did not actually exist. To avoid this problem, all rows containing NaN values was removed before proceeding:

2 Materials and Methods

```
e1_100kb.dropna(inplace=True)
```

This step ensured the sign changes in the eigenvector reflect only genuine compartment transitions.

After cleaning, the function calculated genomic coordinates by assigning each bin a start and end position based on its index and the fixed bin size of 100 kb. With these coordinates in place, it determined the sign of each eigenvector value and assigns segment IDs whenever the sign changes, so that continuous stretches of positive or negative values are grouped into compartments.

Each compartment was then summarized by calculating the mean and sum of its eigenvector values and recording the minimum start and maximum end positions. Segments with a positive mean were classified as compartment A, while those with a negative mean are classified as compartment B. To prevent artifacts caused by missing bins, the function also checks segment borders to ensure that gaps do not create artificial extents.

The function produced two outputs. The first was a tidy data frame of compartments that includes chromosome, start and end positions, mean eigenvector value, segment ID, sign, and compartment label. The second was a sorted list of all unique segment boundaries, which represented the genomic positions of compartment transitions.

To separate the data into A- and B-compartments, a simple filtering step was used. For each genomic interval, the filter compared the compartment label (A or B) with the sign of its position relative to the nearest transition. Intervals that matched the expected direction were collected into one set, while those that fell on the opposite side were collected into another. In practice, this was done with the following code:

```
A_val = result[
    ((result['comp'] == 'A') & (result['start'] < 0)) |
    ((result['comp'] == 'B') & (result['start'] > 0))
].copy()
B_val = result[
    ((result['comp'] == 'A') & (result['start'] > 0)) |
    ((result['comp'] == 'B') & (result['start'] < 0))
].copy()
```

This logic ensured that the data was consistently separated into values supporting the A-compartment and values supporting the B-compartment, based on both the compartment label and the position relative to the transition point.

To verify that our compartment analysis was not biased by compartment assignment, we performed a negative control experiment. Normally, the data are separated into A- or B-compartments (as described in the methods section). In the control analysis, however, the data were instead partitioned according to compartment transitions — whether regions lay on an

A to B or B to A boundary. Because Hi-C data reflect DNA contacts independently of strand orientation, no systematic difference was expected between the two directions.

As an additional safeguard, the same experiment was repeated for CpG islands, repeats and GC content. Filtering was carried out with the following code snippet, which reverses the logic of the standard A/B assignment:

```
A_val = result[
    ((result['comp'] == 'A') & (result['start'] < 0)) |
    ((result['comp'] == 'A') & (result['start'] > 0))
].copy()
B_val = result[
    ((result['comp'] == 'B') & (result['start'] > 0)) |
    ((result['comp'] == 'B') & (result['start'] < 0))
].copy()
```

2.7 Distance between the edges

To evaluate the availability of data across genomic distances and mitigate potential bias in downstream analyses, the reverse cumulative counts of compartment edge half-distances was calculated. First, edges were identified at compartment transitions between A and B (in either direction). For each consecutive pair of such edges, we measured the genomic distance between their start positions and divided it by two, yielding the half-distance between edges.

The distances has then half and computed the reverse cumulative count for each bin, which represented the number of edges with half-distances greater than or equal to that bin. This approach effectively measures the number of edges capable of contributing data at a given spatial scale. By using this metric, it allowed us to normalize other measurements by the available data density and reducing the risk that apparent trends were driven by varying numbers of contributing edges at different distances rather than true biological effects. Counts were calculated up to 2 Mb to encompass the maximum observed separation.

2.8 Error bars calculation

To estimate uncertainty around the binned values in the plot, 95% confidence intervals (CIs) were calculated using a normal approximation to the binomial distribution. For each bin, the standard deviation was computed as:

$$SD = \sqrt{\frac{c}{n}}$$

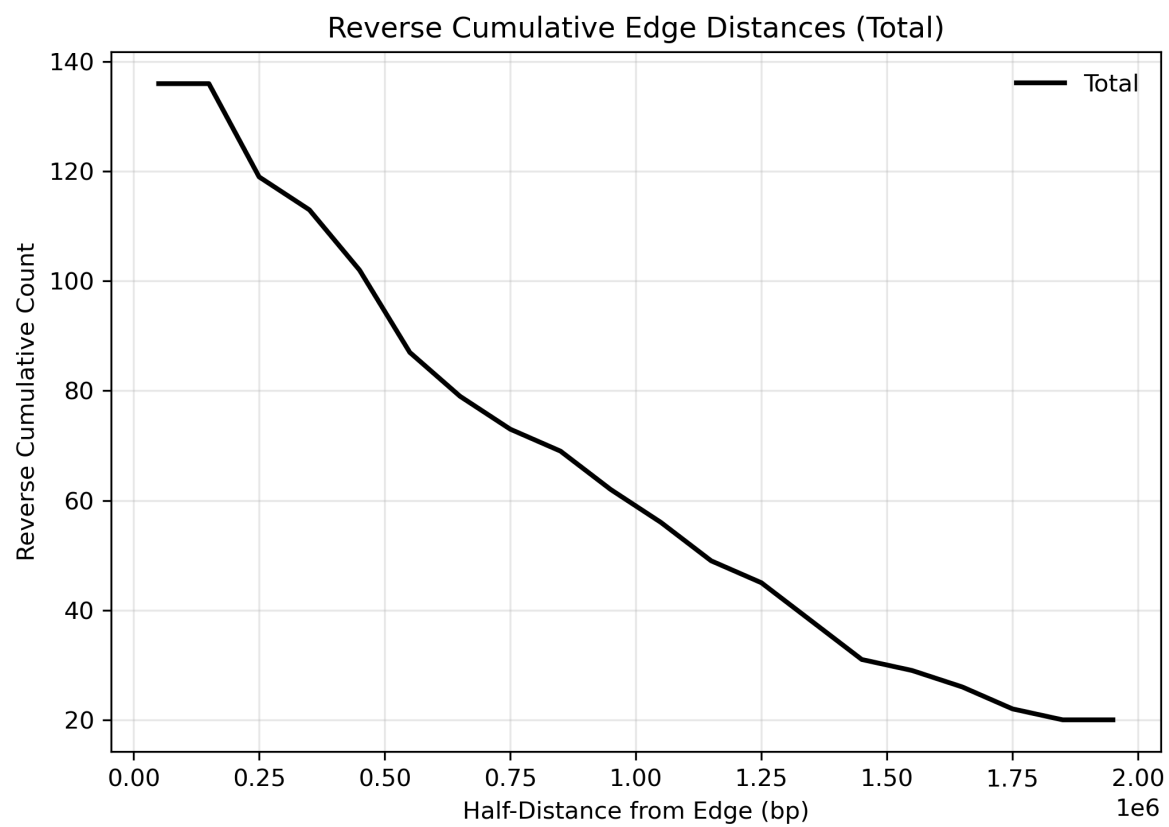


Figure 2.3: Reverse cumulative counts of compartment edge half-distances..

2.8 Error bars calculation

where c is the observed count of events (e.g., repeat instances), and n is the number of positions (or sites) that could potentially contribute to that bin, derived from a reverse cumulative count. To avoid division by zero, a minimum denominator of 1 was enforced.

The error bars represent 95% confidence intervals, calculated as:

$$CI_{95} = 1.96 \times SD$$

This approach accounts for the decreasing number of possible sites at increasing distances from the reference point and provides a conservative estimate of variation across bins.

3 Result

3.1 Aligning GC content to edges in Macaca

The result of compared GC content between A- and B- compartments for chromosome 8 and chromosome X across five Macaca cell types: fibroblasts, sperm, round spermatids, pachytene spermatocytes, and spermatogonia. Chromatin compartments were defined from A to B and B to A structural transitions, and GC content was measured separately for each compartment.

For chromosome 8, the A-compartment consistently showed higher GC content than the B-compartment in all cell types, which aligns with expected genomic organization. In chromosome X, however, the differences between compartments were smaller. In round spermatids, GC content was nearly identical between A- and B-compartments, while in pachytene spermatocytes, spermatogonia, and sperm, A and B-compartments had closely overlapping GC values. Interestingly, in fibroblasts the pattern was reversed, with the B-compartment displaying higher GC content than the A-compartment see Figure 3.1.

3.2 CPG islands

To investigate chromatin dynamics during spermatogenesis in Macaca, we genome-wide changes in accessibility across successive developmental stages. We focused on genomic regions undergoing transitions from open to closed chromatin states and vice versa, as these sites represent potential regulatory hotspots that may contribute to germ cell differentiation. To explore the relationship between these transitions and CpG-rich elements, we obtained a genome-wide track of CpG islands and mapped their positions relative to the transition regions. Each transition was further annotated according to its placement within A- or B-compartments, with compartment A typically associated with transcriptionally active euchromatin and B-compartment corresponding to transcriptionally repressed heterochromatin. By integrating chromatin transitions, CpG island mapping, and compartmentalization, we generated a stage-specific view of genome organization across spermatogenesis.

Figure 3.2 illustrates the distribution of CpG island-associated transitions in fibroblasts on chromosome 8. At distances up to ~0.5 Mb, transition counts were comparable between A- and B-compartments, but beyond this range the A-compartment exhibited substantially higher counts. This enrichment persisted until ~1.5 Mb, after which both compartments declined toward zero. The drop at larger distances likely reflects technical constraints: to measure a

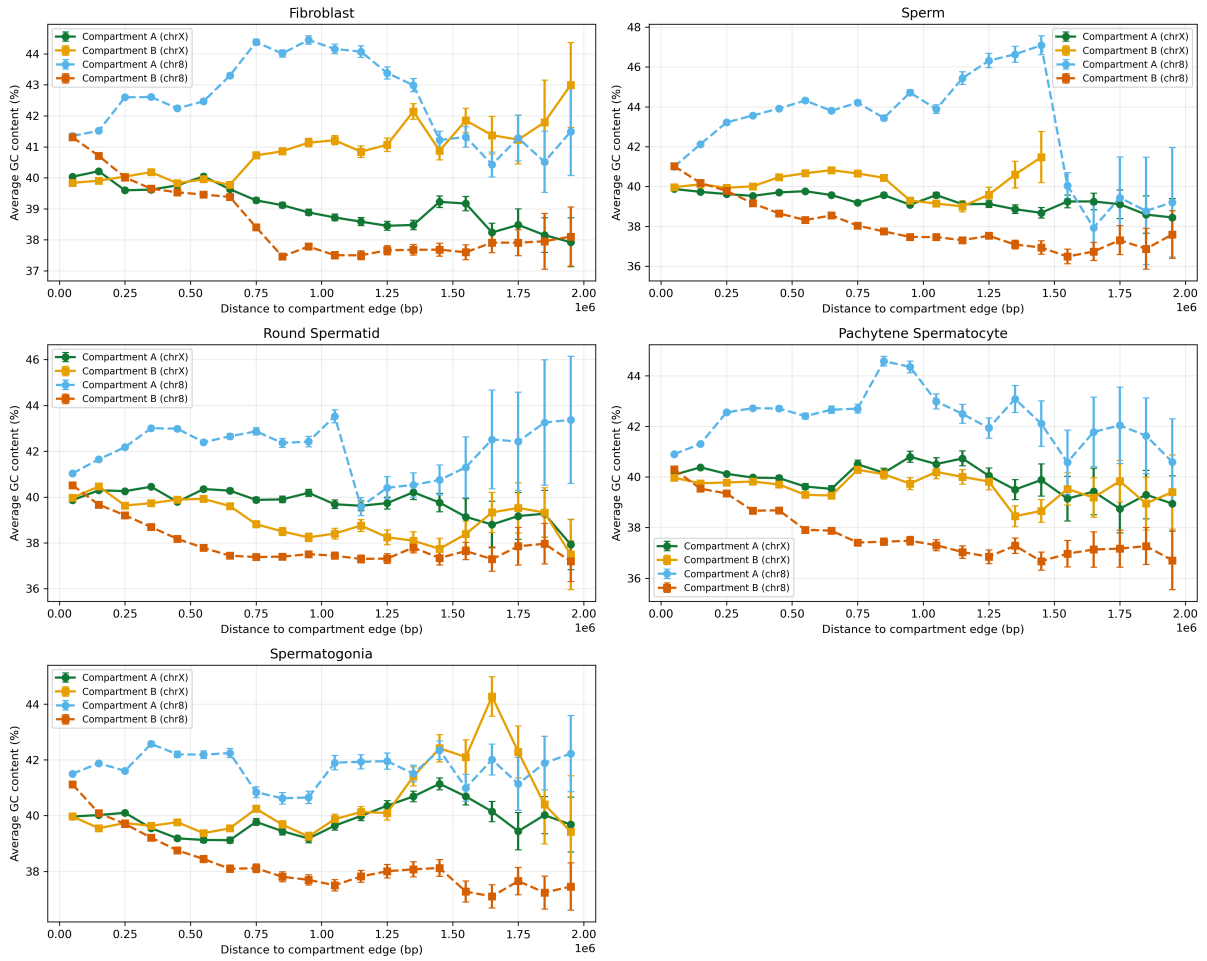


Figure 3.1: Compartment analysis for chromosomes X and 8 in five *Macaca* cell types—fibroblasts, sperm, round spermatids, pachytene spermatocytes, and spermatogonia. GC content was remapped to genomic locations where chromatin structure transitions between compartments A and B. The data were separated so that each compartment contains only A or B structure. For each genomic bin, the plot shows the mean GC content, with error bars representing the standard deviation of the mean, adjusted by the effective edge contribution (accounting for how many edges contribute at longer distances). For chromosome 8, the A-compartment exhibits higher GC content than the B-compartment. In tissues involved in spermatogenesis, chromosome X shows similar GC content in both A- and B-compartments. In fibroblasts, GC content is higher in the B-compartment than in the A-compartment.

3 Result

transition at 1.5 Mb, the minimum spacing between sites must be 3 Mb, which is relatively rare across the genome. This limitation is evident even though counts were normalized to account for the number of possible contributing sites. By contrast, chromosome X showed no compartment-specific differences in fibroblasts, with A- and B-compartments displaying similar counts across all distances.

In spermatogonia, no compartment-specific differences were observed on either chromosome 8 or chromosome X. Transition counts remained comparable between A- and B-compartments across all distances, in contrast to the fibroblast profile where chromosome 8 showed A-compartment enrichment.

In pachytene spermatocytes, chromosome 8 displayed a distinct compartmental pattern. Between 0 and ~1.25 Mb, transition counts were higher in the A-compartment compared to the B-compartment, but at longer distances the two converged. On chromosome X, however, A- and B-compartments remained similar, showing no evidence of compartment-specific enrichment.

In round spermatids, chromosome 8 again showed a bias toward the A-compartment, with more transitions observed from 0 to ~1.1 Mb. A comparable, though slightly shifted, pattern was seen on chromosome X, where enrichment in the A-compartment was detected between ~0.25 and 1.3 Mb.

Finally, in mature sperm, chromosome 8 exhibited consistent A-compartment enrichment from 0 to ~1.5 Mb. On chromosome X, the effect was more modest, with only a slight increase in A-compartment counts between ~1.0 and 1.75 Mb.

Together, these results reveal stage-specific differences in the relationship between chromatin transitions, CpG islands, and genome compartmentalization. Fibroblasts and spermatogonia lack strong compartmental biases, while later spermatogenic stages, particularly pachytene spermatocytes, round spermatids, and mature sperm, display pronounced enrichment of CpG island-associated transitions in the A-compartment, especially on chromosome 8.

3.3 Simple repeat

We next examined the relationship between chromatin transitions and simple repeats using the same strategy as for CpG islands. The genomic positions of simple repeats were remapped relative to transition sites, and the counts in each distance bin were normalized by the number of transition sites that could contribute to that bin. This approach allowed us to determine whether simple repeats are enriched near transition regions and whether their distribution differs between A and B-compartments.

In fibroblasts, chromosome 8 showed a slight enrichment of simple repeats in the B-compartment compared to the A-compartment. On chromosome X, however, the opposite pattern was observed, with more simple repeats detected in the A-compartment. Across both chromosomes, simple repeats were generally more abundant close to transition sites, with their frequency declining as distance increased.

3.3 Simple repeat



Figure 3.2: **Stage-specific enrichment of CpG island-associated chromatin transitions across A/B compartments in *Macaca*.**

The figure displays chromatin transitions align to CpG islands across developmental stages of spermatogenesis. Plots are arranged in two columns: chromosome 8 (left) and chromosome X (right). Rows correspond to cell types, from top to bottom: fibroblasts, spermatogonia, pachytene spermatocytes, round spermatids, and mature sperm. On chromosome 8, compartment A transitions are shown in green and compartment B in red. On chromosome X, compartment A is represented in purple and compartment B in blue. The figure highlights gradual, stage-dependent variation in the compartmental distribution of CpG-associated transitions, with a tendency toward increased A-compartment representation during the later stages of spermatogenesis—particularly on chromosome 8.

3 Result

In spermatogonia, chromosome 8 did not display any clear compartment-specific differences, as simple repeat counts were comparable between A- and B-compartments across all distances. On chromosome X, simple repeats were evenly distributed between compartments up to ~0.5 Mb, but from ~0.5 to 2 Mb the A-compartment contained higher counts. Similar to fibroblasts, spermatogonia also showed strong enrichment of simple repeats near transition sites, with counts decreasing at greater distances.

In pachytene spermatocytes, chromosome 8 again showed more simple repeats in the B-compartment than in the A-compartment, with a strong enrichment close to transition sites. On chromosome X, an A-compartment bias emerged beyond ~0.75 Mb, where more simple repeats were observed in A compared to B.

In round spermatids, chromosome 8 displayed a consistent bias toward the B-compartment, with more simple repeats present there than in the A-compartment. By contrast, chromosome X showed the opposite trend, with higher counts of simple repeats in the A-compartment.

Finally, in mature sperm, chromosome 8 continued to show an enrichment of simple repeats in the B-compartment, while chromosome X again displayed more simple repeats in the A-compartment.

3.4 Nested repeat

We also analyzed the distribution of nested repeats relative to chromatin transitions using the same mapping and normalization strategy applied to CpG islands and simple repeats. The results for nested repeats closely paralleled those observed for simple repeats.

In fibroblasts, chromosome 8 showed more nested repeats in the B-compartment, while chromosome X displayed a bias toward the A-compartment. In spermatogonia, chromosome 8 showed no compartment-specific differences, whereas chromosome X exhibited an A-compartment enrichment beyond ~0.5 Mb. Pachytene spermatocytes, round spermatids, and mature sperm followed the same overall trends: chromosome 8 consistently contained more repeats in the B-compartment, while chromosome X showed enrichment in the A-compartment. Across all stages and both chromosomes, nested repeats were most abundant near transition sites, with counts decreasing at larger distances.

3.5 Control to with A- and B-compartment

The outcomes of the control analysis for simple and nested repeats are presented in Figures Figure 3.5 and Figure S6.1. Across all tissues, the distributions for both chromosome 8 and chromosome X show identical counts within each bin.

This consistency indicates that the choice of chromosome split does not introduce a biological difference in the observed patterns. In other words, the apparent symmetry between the A to

3.5 Control to with A- and B-compartment

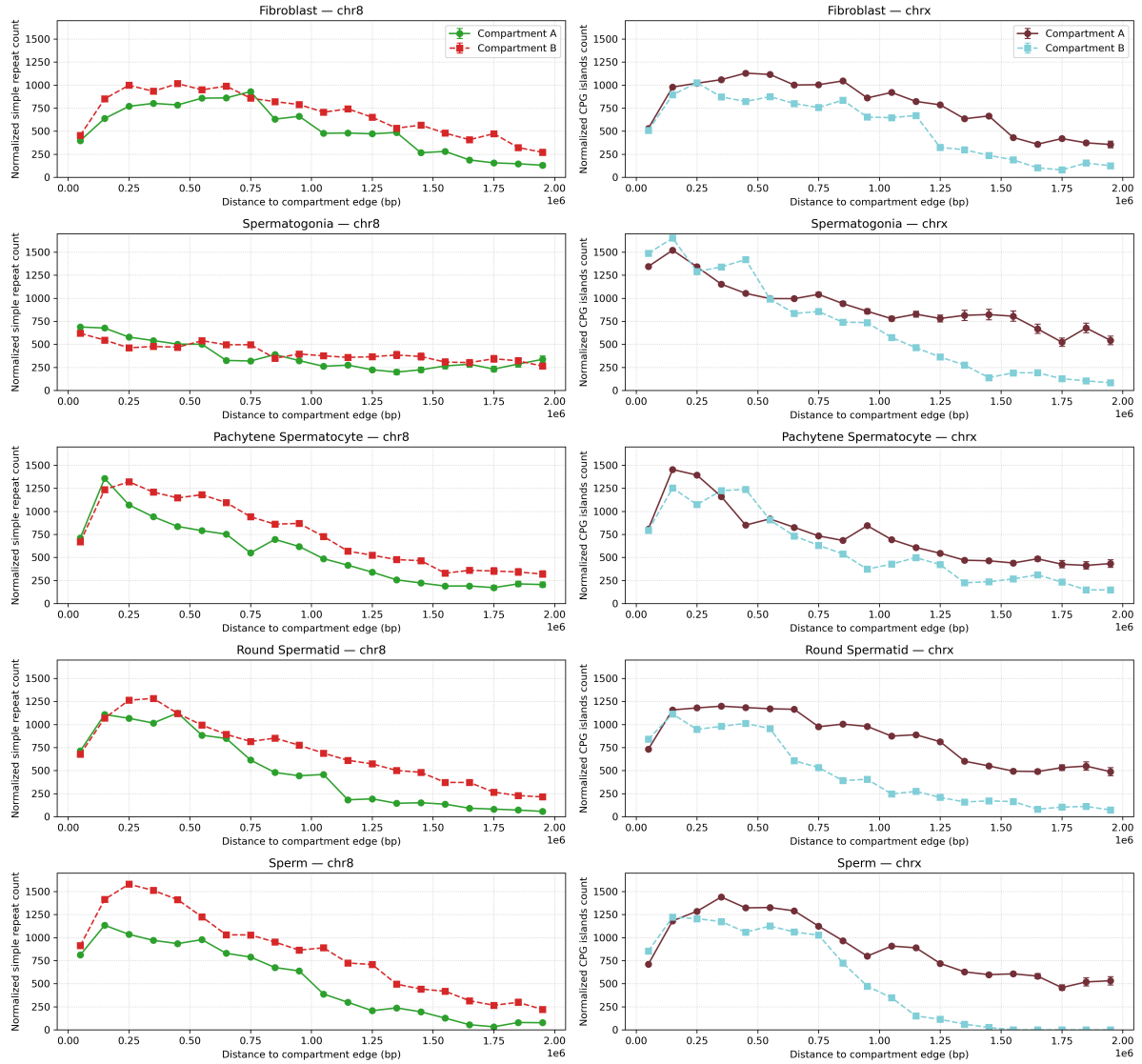


Figure 3.3: **Distribution of simple repeats relative to chromatin transition sites in fibroblasts and spermatogenic stages.**

Chromosome 8 is shown on the left and chromosome X on the right. Rows represent different cell types: fibroblasts, spermatogonia, pachytene spermatocytes, round spermatids, and mature sperm. In each panel, simple repeat counts were mapped relative to transition sites, and values in each distance bin were normalized by the number of transition sites that could contribute to that bin.

3 Result

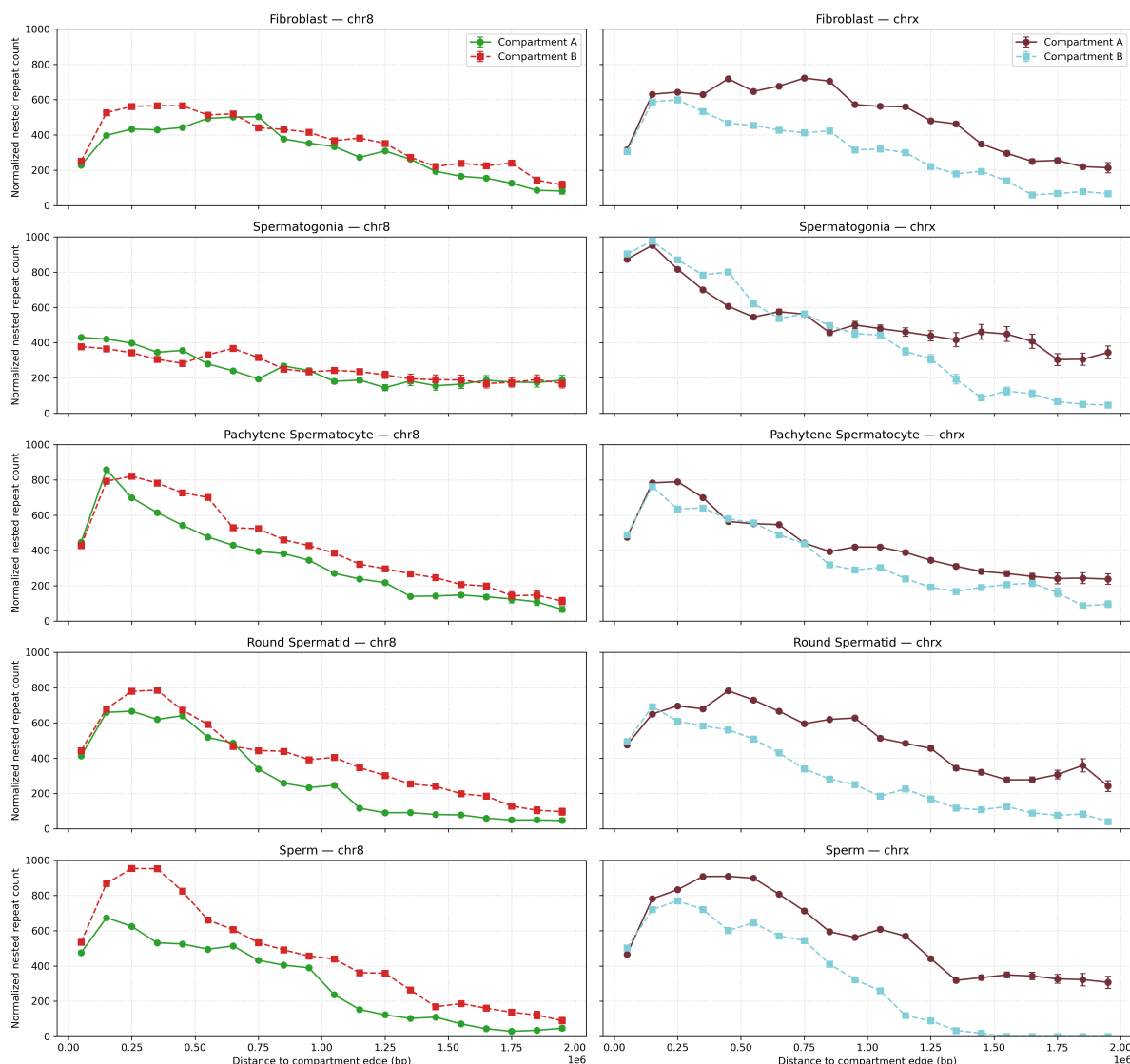


Figure 3.4: **Distribution of nested repeats relative to chromatin transition sites in fibroblasts and spermatogenic stages.**

chromosome 8 can be seen on graphs to the left, and chromosome x can be seen on graphs to the right. Rows represent different cell types: fibroblasts, spermatogonia, pachytene spermatocytes, round spermatids, and mature sperm. In each panel, nested repeat counts were mapped relative to transition sites, and values in each distance bin were normalized by the number of transition sites that could contribute to that bin.

3.5 Control to with A- and B-compartment

B and B to A transitions confirms that the results are not biased by the direction of the split, validating the robustness of the control setup.

When running the control analysis with GC content, we plotted all five tissues for both chromosome 8 and chromosome X (Figure Figure 3.6).

For chromosome 8, no significant differences were observed between the A to B and B to A splits across tissues, with the exception of spermatogonia, where a deviation appeared in the region spanning 0.25 Mb to 1.25 Mb.

For chromosome X, the results varied by tissue. In fibroblasts, the B to A split showed consistently higher GC content compared to A to B within the 0–1 Mb region. A similar pattern was observed in sperm, where GC content remained higher in B to A. By contrast, in round spermatids and pachytene spermatocytes, there were no significant differences between the two splits. In spermatogonia, however, GC content was again higher in B to A compared to A to B.

The resulting profiles in Figure Figure 3.7 illustrate the distribution of CpG islands relative to compartment transitions (A to B vs. B to A) on chromosomes 8 and X across multiple cell types.

On chromosome 8, fibroblasts and sperm show a pronounced asymmetry: between ~0.5 and 1.5 Mb from transition edges, CpG islands are more abundant in regions undergoing A to B transitions than in those switching from B to A. By contrast, in spermatogonia and pachytene spermatocytes, CpG islands display a more balanced distribution, with comparable counts in both transition directions across the full range of distances examined.

On chromosome X, no strong directional bias is detected. Across all tissues analyzed—including fibroblasts, spermatogonia, pachytene spermatocytes, round spermatids, and sperm—the counts of CpG islands at A to B and B to A transitions remain similar across the 0–2 Mb distance range. This uniformity suggests that, unlike chromosome 8, the X chromosome does not exhibit a consistent asymmetry in the distribution of CpG islands relative to compartment switching.

Together, these results indicate that CpG islands are not randomly associated with chromatin transitions but instead show chromosome- and stage-specific patterns. Chromosome 8 exhibits directional enrichment of CpG islands, particularly in fibroblasts, sperm, and round spermatids, while the X chromosome appears more symmetric, lacking a consistent bias between A to B and B to A transitions.

3 Result

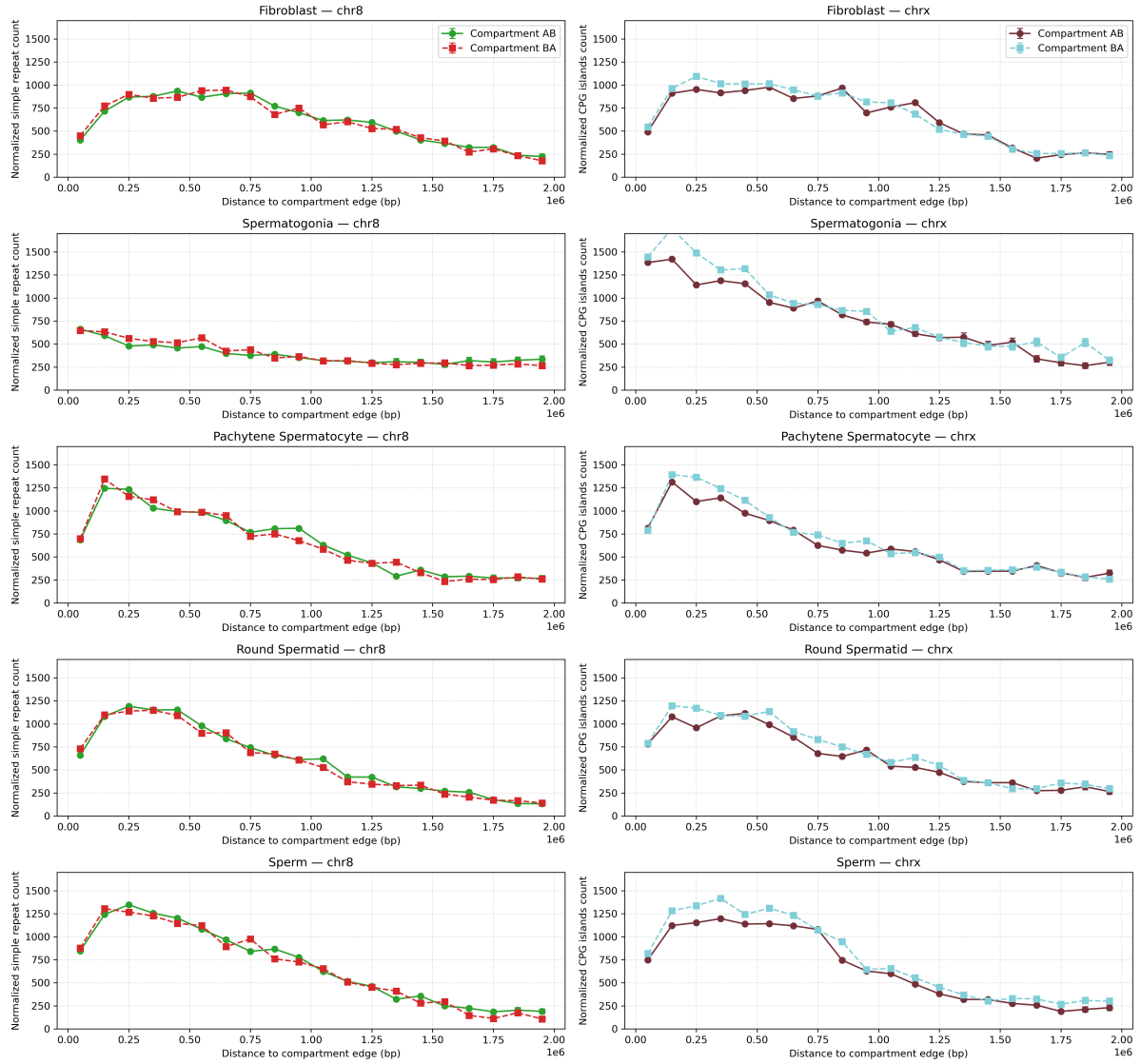


Figure 3.5: Control analysis of simple repeats. The distributions for chromosome 8 and chromosome X show identical counts across bins in all tissues, indicating no biological difference introduced by this chromosome split.

3.5 Control to with A- and B-compartment

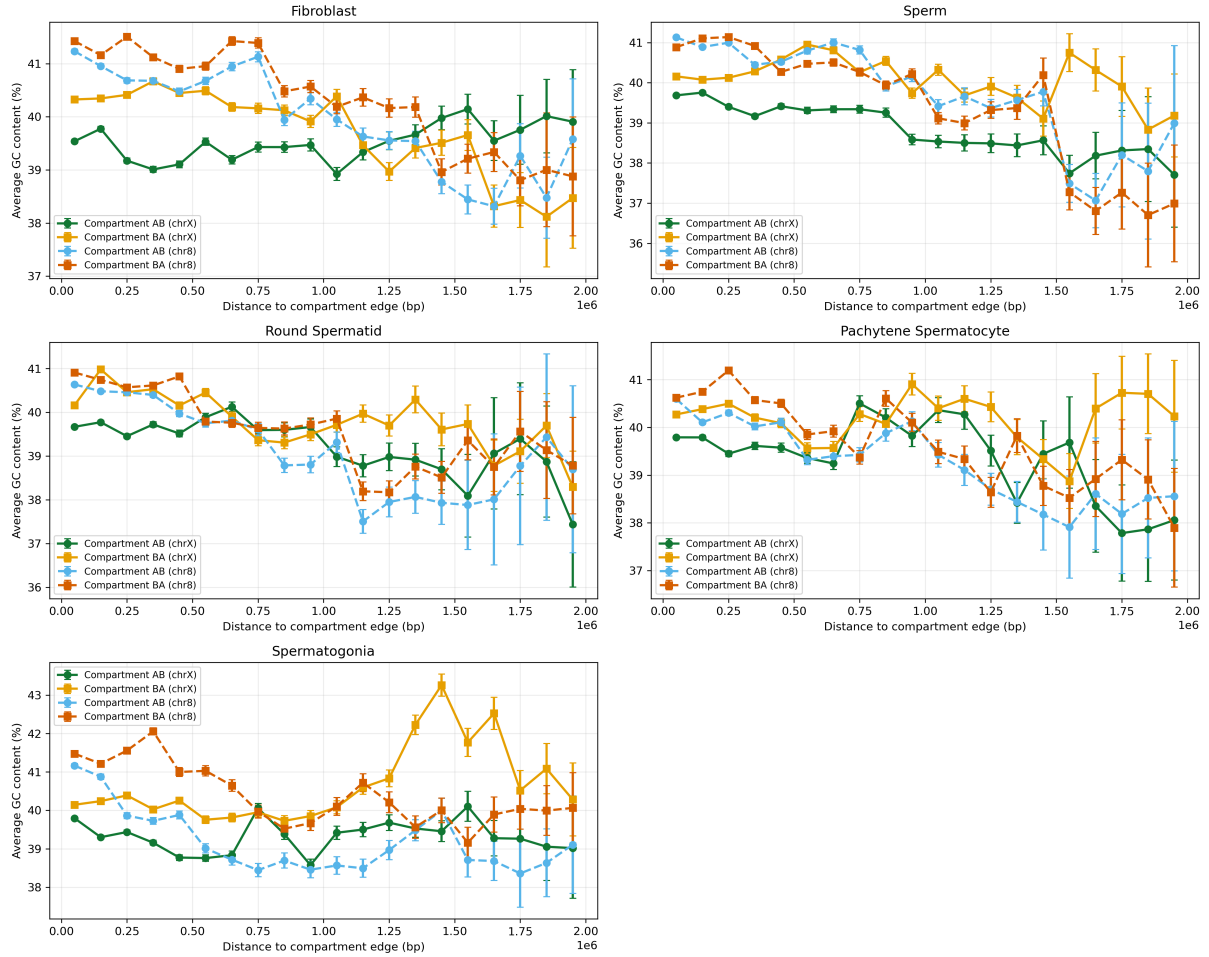
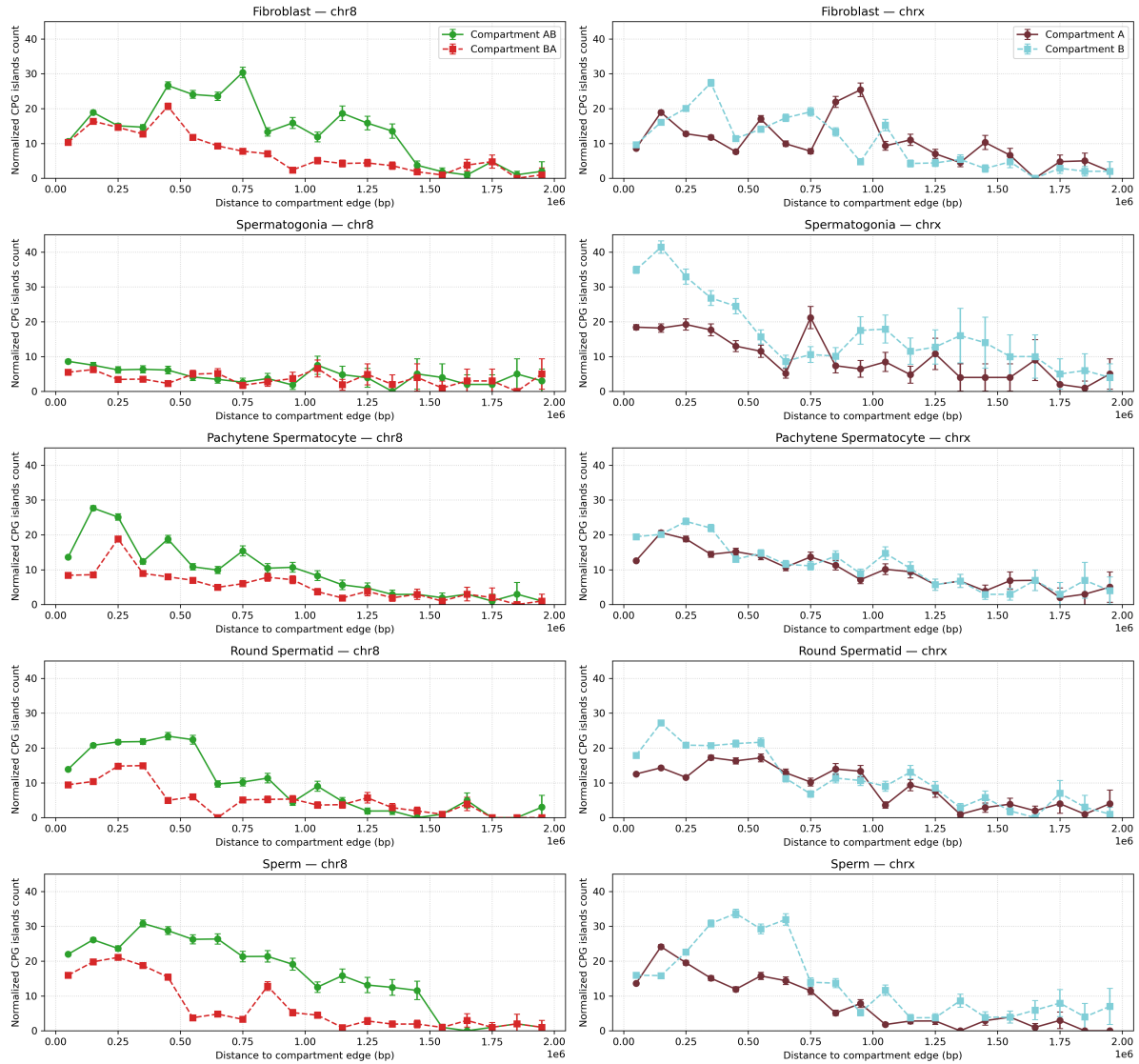


Figure 3.6: Control analysis of GC content for chromosomes 8 and X across five tissues. For chromosome 8, no significant differences between A to B and B to A were detected except in spermatogonia (0.25–1.25 Mb). For chromosome X, higher GC content was observed in B to A compared to A to B in fibroblasts (0–1 Mb), sperm (0–1 Mb), and spermatogonia, while round spermatids and pachytene spermatocytes showed no differences.

3 Result



Negative control analysis of CpG islands on chromosome 8 across five macaque cell types (fibroblast, spermatogonia, pachytene spermatocyte, round spermatid, and sperm). A to B (AB) and B to A (BA) compartment transitions were compared.

Negative control analysis of CpG island proximity to A/B compartment transitions on chromosome 8 across spermatogenic stages in *Macaca*.

This control analysis compares CpG island frequencies near directional compartment transitions—specifically transitions from A to B (AB) and from B to A (BA) rather than focusing on static A or B compartments, as in the main analysis. For each of the five cell types (fibroblast, spermatogonia, pachytene spermatocyte, round spermatid, and mature sperm), genomic regions undergoing compartment status changes were identified, and CpG island counts were measured relative to these transition edges.

Plots follow the same structure as in the main figure: chromosome 8 (left column) and chromosome X (right column), with rows arranged from fibroblast (top) to sperm (bottom). On chromosome 8, Ab to B transitions are shown in green and B to A in red; on chromosome X, A-to-B is shown in purple and B-to-A in blue.

30

Figure 3.7: In fibroblasts on chromosome 8, higher CpG island counts were observed in the A to B category between ~0.8 and 1.4 Mb. Similar differences between AB and BA transitions were also present in pachytene spermatocytes, round spermatids, and sperm, particularly from 0 to ~1.2 Mb. On chromosome X, CpG island counts for AB and BA transitions were comparable across all stages.

4 Dissusion

4.1 GC content and A/B compartmentalization on the macaque X chromosome

As earlier established the GC content is a strong methode for identifying the A/B compertments, with A-compartments being GC rich and B-compartments being GC poor. The analysis done in this paper of the macaque chromosome 8 shows the same realtionship in all five cell types examined, as seen on Figure 3.1. In contrast, chromosome X departs from this pattern in several ways. First, the precetags diffrence of GC content between A and B is reduced in spermatogonia, pachytene spermatocytes, round spermatids, and sperm. Second, fibroblasts show an inversion higher GC in B-compartment than in A-compartment. Despite this relationship seen in chromosome 8, it is important to have in mind, that this approach has three main limitations: (i) edge positions are localized only to 100 kb, so features within $\sim\pm 50$ kb of a edges may be blurred by binning; (ii) the amount of data points declines with increasing distance due to many of the edges close proximyty to each other, which widens confidence intervals at longer distances even after normalization; and lastly (iii) because PC1 was phased using GC to orient A/B compartments.

The chromosome 8 follows the classic GC A/B relationship while chromosome X does not highlights an important distinction: GC content is a robust predictor of compartment state on autosomes but not necessarily on sex chromosomes. The attenuated contrast and apparent fibroblast reversal suggest that on chromosome X, base composition is a strong prior for compartmentalization but not a determinant under all conditions. Regulatory programs such as dosage compensation, sex-specific chromatin environments, and the unusual sequence architecture of the X chromosome can override the typical GC to A mapping, compressing or inverting the signal.

These findings argue for additional validation when interpreting compartment states on chrX. Cross-referencing with independent chromatin marks and stratifying X into domain types (e.g., ampliconic versus non-ampliconic, CGI-rich versus CGI-poor) would help clarify whether the reduced GC-compartment correlation reflects biological mechanisms or labeling artifacts. More broadly, the results underscore that while GC content is a useful baseline for compartment identity, its predictive power may be context-dependent—particularly on the sex chromosomes, which are subject to distinct evolutionary and regulatory pressures.

4.2 CpG islands and chromatin transitions during spermatogenesis

(this is kinda shit) Our analysis reveals that CpG island-associated chromatin transitions show distinct, stage-specific, and chromosome-specific patterns across macaque spermatogenesis. On chromosome 8, fibroblasts already exhibit a strong enrichment of CpG island-linked transitions in the A-compartment, particularly at distances up to ~1.5 Mb. In spermatogonia, by contrast, transitions are distributed more evenly between compartments, with little evidence of bias. At later germline stages—pachytene spermatocytes, round spermatids, and mature sperm—an A-compartment bias re-emerges, indicating that CpG islands become increasingly associated with open, transcriptionally active regions as chromatin undergoes extensive remodeling during meiosis and spermiogenesis. This developmental trajectory suggests that the relationship between CpG islands and compartment identity is not fixed but dynamically regulated, absent in early germline precursors, present in differentiated somatic fibroblasts, and reinforced during the major reorganizational phases of germ cell maturation.

(this is okay) By contrast, the X chromosome generally shows weaker and more variable compartmental differences. In fibroblasts and spermatogonia, no enrichment is observed, mirroring the pattern seen on chromosome 8. In round spermatids, however, a modest A-compartment bias emerges, and in sperm this trend persists but remains less pronounced than on the autosome. These findings align with the idea that the X chromosome is subject to distinct regulatory pressures—such as meiotic sex chromosome inactivation (MSCI) and post-meiotic repression—that may dampen or delay the coupling between CpG islands and active chromatin states. The X appears less responsive to the same sequence-based cues that strongly partition CpG island-associated transitions on autosomes.

CpG islands are well known to act as promoters, boundary elements, and nucleation sites for chromatin remodeling. Their enrichment within A-compartments during later spermatogenic stages suggests that they may stabilize or reinforce transcriptionally permissive domains at precisely the time when the 3D genome undergoes wholesale remodeling. In particular, pachytene spermatocytes and spermatids show both loss of canonical TADs and emergence of refined compartmental structures. CpG islands could therefore help define or insulate these reorganized domains, ensuring that key regulatory programs for gamete development are maintained.

Several factors caution against over-interpretation. First, transition edges are identified at 100-kb resolution, which limits positional accuracy; associations between CpG islands and compartments may therefore be blurred within ± 50 kb of boundaries. Second, the normalization procedure compensates for distance-dependent availability of transitions, but the steep decline in counts at longer distances still reflects limited genomic sampling. Third, our analysis does not distinguish between CpG islands at active promoters, intergenic CGIs, or those embedded within repeats—categories that may behave differently. Finally, on the X chromosome, compartment labels are phased by GC content, raising the possibility of mislabeling in contexts where the GC-compartment correlation is weak.

4.3 Discussion: Simple repeats and compartment-specific enrichment

Our analysis of simple repeats reveals consistent stage- and chromosome-specific differences in their relationship to chromatin compartments. On chromosome 8, simple repeats are preferentially associated with the B-compartment in fibroblasts, pachytene spermatocytes, round spermatids, and mature sperm, with spermatogonia representing the only stage without a clear bias. By contrast, the X chromosome shows the opposite pattern: simple repeats are generally enriched in the A-compartment across most developmental stages, with only minor differences in spermatogonia at shorter distances. This reciprocal behavior between the autosome and the sex chromosome highlights that the genomic context strongly shapes how repetitive elements interact with higher-order chromatin structure.

A second consistent feature across all stages and both chromosomes is the sharp enrichment of simple repeats near chromatin transition sites, with counts declining steadily at larger distances. This pattern suggests that simple repeats may preferentially accumulate at or around regions where chromatin states switch between active and inactive configurations. Whether repeats directly contribute to establishing such boundaries, or instead persist in these regions due to reduced selective constraints, remains unclear. Nonetheless, their proximity to transition sites supports the idea that repetitive elements may play a structural role in shaping domain organization, either by altering nucleosome positioning, promoting local heterochromatinization, or acting as anchors for chromatin remodeling complexes.

The contrasting biases between chromosome 8 and the X chromosome may reflect differences in the sequence composition, regulatory environment, and evolutionary dynamics of these genomic regions. On autosomes, simple repeats appear to reinforce repressed, heterochromatic domains, while on the X they are more often linked to active A-compartments. One possible explanation is that the unique regulatory pressures on the X chromosome—such as dosage compensation, sex-specific silencing, and its repeat-rich architecture—decouple the typical relationship between repeats and heterochromatin, allowing repeats to persist or even become enriched in euchromatic environments.

Together, these findings suggest that simple repeats are not randomly distributed with respect to 3D genome organization, but instead exhibit compartment- and chromosome-specific associations that shift across developmental stages. Their enrichment near transition regions in particular points to a potential role in modulating or stabilizing chromatin boundaries during spermatogenesis. Further work, integrating repeat subtypes, transcriptional data, and histone modification profiles, will be needed to test whether these associations are structural, regulatory, or a byproduct of local sequence composition.

((This suggests that CpG islands are preferentially associated with loci moving from active to repressive chromatin environments in these contexts. In round spermatids, a similar enrichment is observed closer to the transition site (0–1 Mb), again with higher counts in A to B compared to B to A))

4.4 Control analysis of A to B vs. B to A transitions

The negative control analysis was designed to test whether directional compartment assignments (A to B versus B to A) could introduce spurious structure into our results. Since Hi-C captures contact frequencies independently of strand orientation, the null expectation is symmetry: features such as repeats, GC content, or CpG islands should be distributed equivalently across both types of transitions. In large part, our findings support this expectation. For example, simple and nested repeats showed no systematic differences across transition categories or tissues, confirming that our splitting strategy did not artificially bias these features. Similarly, on the X chromosome, CpG islands and GC content exhibited balanced distributions across A to B and B to A transitions in most tissues.

However, several deviations from symmetry were observed. Most notably, chromosome 8 displayed stage-specific enrichment of CpG islands in A to B transitions, particularly in fibroblasts, sperm, and round spermatids. Likewise, GC content differences emerged on chromosome X in some tissues, with B to A regions showing consistently elevated values. These patterns raise an important interpretive challenge: are they biologically meaningful, or could they reflect artifacts introduced by the partitioning scheme itself?

One possible explanation is that the asymmetries arise from the structure of the data splitting procedure. By conditioning on compartment boundaries, the analysis inherently emphasizes regions undergoing dynamic changes in chromatin state. If features such as CpG islands or GC-rich domains are unevenly distributed relative to those boundaries, then separating regions into A to B versus B to A could exaggerate apparent differences even if no true directional bias exists. In other words, what appears to be a biological signal may instead reflect the geometry of the compartment transition and how bins are aligned across the genome.

At the same time, the tissue- and chromosome-specific nature of these asymmetries argues against a purely technical artifact. If the differences were driven solely by the method of splitting, one would expect them to occur broadly and consistently across tissues and chromosomes. Instead, the enrichment of CpG islands in A to B regions is largely confined to chromosome 8 and restricted to particular developmental stages, suggesting that at least part of the signal reflects genuine biology. This interpretation is supported by the absence of comparable asymmetries on the X chromosome, which serves as an internal control.

Taken together, these results highlight both the strength and the limitation of the control design. On the one hand, the null expectation of symmetry holds for many features, providing reassurance that our compartmental analyses are not systematically biased. On the other hand, the exceptions underscore that care must be taken when interpreting direction-specific enrichments. Some patterns, particularly those restricted to individual chromosomes and developmental stages, may represent true biological processes—for instance, preferential positioning of CpG islands at A to B transitions during chromatin remodeling. Others, however, may be artifacts of the binning and splitting procedure. Distinguishing between these possibilities will require further analyses, such as permuted controls, higher-resolution maps, or independent validation datasets.

4.5 Reproduction Attempt of GPN

maybe

Initial attempts to apply the workflow to a baboon genome dataset encountered technical difficulties. These issues were not intrinsic to the baboon data itself but stemmed from the implementation of the workflow. To ensure compatibility with the model and maintain consistency with the original study, the Anatoptis genome—used as the reference in the published work—was selected as a pilot dataset.

Despite these efforts, the model continued to encounter out-of-memory errors during execution. Even with eight GPUs, the memory demands of the GPN model remained prohibitive under the given cluster configuration. These persistent failures highlighted a significant limitation: the model’s architecture and resource demands exceeded the practical limits of the available hardware.

Our attempts to reproduce the Genome-Pretrained Network (GPN) workflow highlighted both the promise of the approach and the challenges of applying it under limited computational resources. While we successfully generated pretraining-compatible datasets and partially executed the workflow, full-scale training was hindered by two main factors: insufficient documentation and substantial hardware requirements.

The original GPN study relied on multiple high-memory GPUs (A100s with 80–100 GB RAM), whereas our available infrastructure consisted of GPUs with less than half this capacity. This mismatch resulted in repeated out-of-memory errors and prevented us from scaling the model to the level reported in the original publication. Although we explored workarounds such as chromosome-level partitioning, these only enabled partial execution and did not allow for comprehensive evaluation.

This outcome underscores a broader limitation of cutting-edge genomics deep learning models: their reliance on specialized hardware makes reproducibility difficult for smaller research groups. At the same time, the successful preparation of compatible datasets and the partial execution steps demonstrate that the workflow is adaptable in principle. Future work should prioritize clearer documentation and optimization strategies that reduce the model’s hardware footprint, thereby lowering barriers to entry for the wider community.

5 Conclusion

Analyses of GC content, CpG islands, and simple repeats relative to A/B compartments and their transitions reveal both stable principles of 3D genome organization and notable exceptions. On autosomes such as chromosome 8, the canonical link between GC-rich, CpG island–dense regions and active A-compartments is maintained across developmental stages, underscoring the reliability of the compartment-calling strategy. In contrast, the macaque X chromosome departs from this pattern, showing attenuated or inverted correlations that reflect its distinctive regulatory environment and evolutionary history.

CpG islands show dynamic, stage-specific associations with compartment transitions, becoming enriched in A to B regions during later stages of spermatogenesis when chromatin is extensively reorganized. Simple repeats, by comparison, display reciprocal biases between autosomes and the X chromosome and accumulate at compartment boundaries, consistent with potential structural or modulatory roles.

Control analyses provide important context for these observations. Symmetry between A to B and B to A transitions holds in many cases, yet selective deviations—such as CpG island enrichment on chromosome 8 and GC asymmetries on the X—raise the possibility that both true biological processes and methodological artifacts are contributing to the signal. Independent validation with orthogonal datasets and higher-resolution maps will be crucial for resolving this ambiguity.

Altogether, the findings indicate that while GC content and CpG islands are strong predictors of compartment identity on autosomes, their predictive power is context-dependent and less reliable on the X chromosome. Sequence composition, repeats, and chromatin structure emerge as deeply interconnected, shaped by both general organizing principles and chromosome-specific constraints. Future work integrating chromatin marks, transcriptional activity, and experimental perturbations will be essential to separate methodological effects from genuine biology and to refine models of how sequence features guide higher-order genome folding.

5.1 futher investigation

This study establishes a foundation for linking sequence features with chromatin compartments during spermatogenesis, but several directions remain open. A key next step is to integrate protein-binding data, particularly CTCF, to test whether compartment edges coincide with architectural anchor points and how these relationships change across developmental stages.

Recombination rates also warrant investigation. Because meiotic hotspots are shaped by chromatin environment, assessing crossover frequency near compartment boundaries could connect 3D genome structure with germline evolution. Structural variants such as inversions provide another opportunity, as they can reposition or disrupt boundaries and reveal causal effects of sequence context.

Finally, higher resolution will be essential. Our current 100-kb compartment calls provide a broad view but obscure finer features. Moving to 10-kb resolution would sharpen boundary localization and enable more precise comparisons with sequence elements, regulatory sites, and recombination maps. Ongoing efforts in this direction within the group will provide an important resource for these future analyses.

6 Supplemental Figures

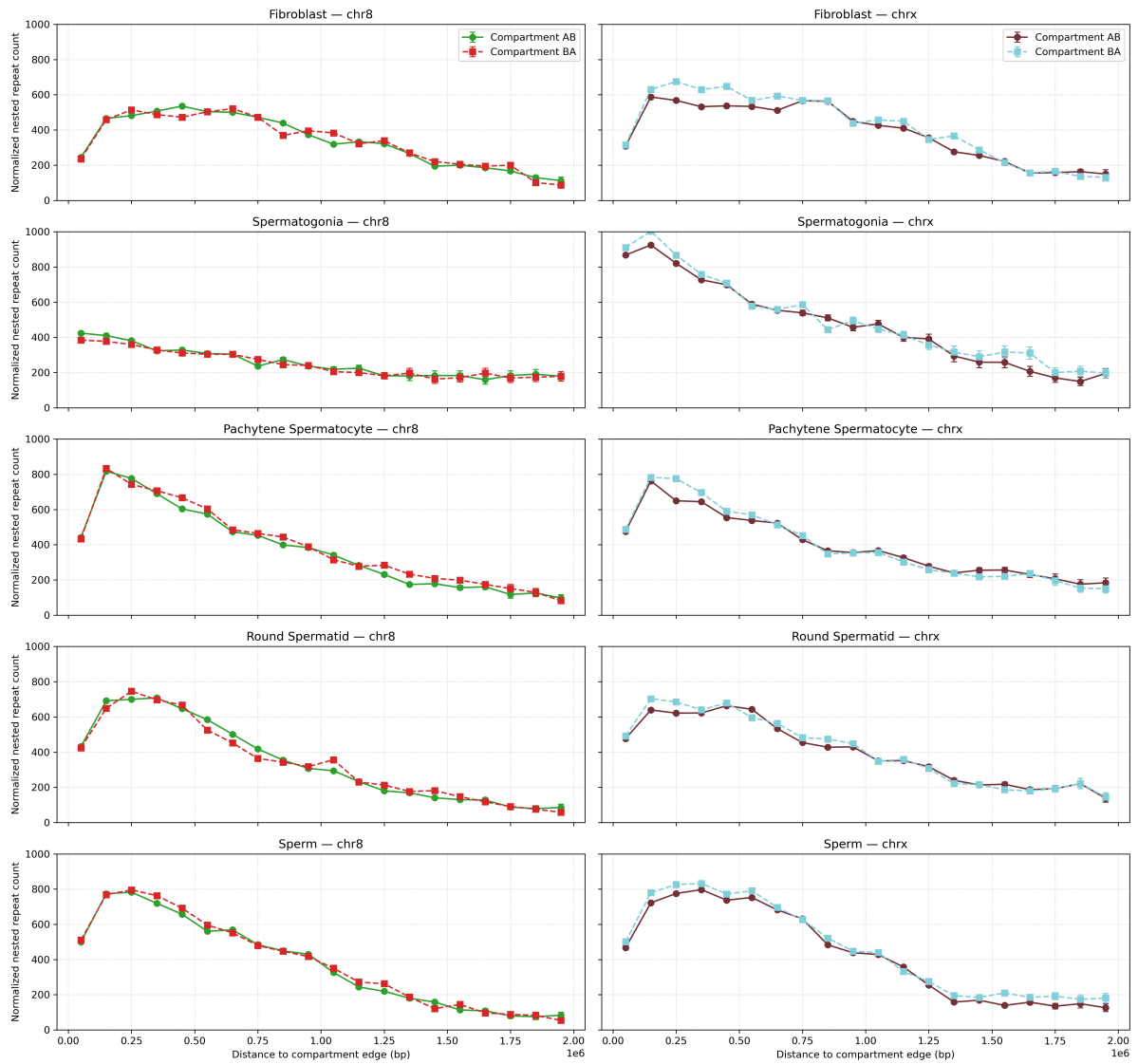


Figure S6.1: **Figure:** GC content but contol

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All computations were performed on GenomeDK (GDK), an HPC cluster located on Aarhus University, and most of the processing of the data was handled by a gwf workflow (GenomeDK 2023), a workflow manager developed at GDK. Developing the manuscript and all associated writing, editing, and rendering the project was also performed on GDK so work seamlessly with the analysis. I would like to thank GDK and Aarhus University for providing computational resources and support that contributed to these research results. None of this would have been possible on my own laptop from 2013.

Use of Generative Artificial Intelligence As per Aarhus University guidelines (IT n.d.), I hereby disclose my use of Generative Artificial Intelligence (GAI) tools during the preparation of this thesis. All usage was conducted critically and responsibly, ensuring that the final work remained my own and adhered to academic integrity standards. These tools supplemented my expertise and accelerated routine tasks, but all creative, analytical, and scientific contributions are my own.

7.3 Copilot

I used Copilot for code completion and debugging, as well as modifying existing code to reflect updated variables and experimental needs.

7.4 ChatGPT

ChatGPT provided syntax support, to accelerate my learning of new programming languages. it also help me with proof reading.

7.5 Cloudi AI

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